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# An In-Depth Analysis of Modern Phishing Detection Methods: A Comprehensive Systematic Literature Review

Doniya Mohamed[1,2] | Mostafa M. El-Gayar[1,3] | Tarek M. Hassan[4] | Mohammed Elmogy[1]

[1]Information Technology Department, Faculty of Computers and Information, Mansoura University, Mansoura, Egypt | [2]Cyber Security Department, Faculty of Artificial Intelligence, Delta University for Science and Technology, Coastal International Road, Gamasa 35712, Egypt | [3]Department of Computer Science, Arab East Colleges, Riyadh 11583, Saudi Arabia | [4]Faculty of Artificial Intelligence, Delta University for Science and Technology, Gamasa 35712, Egypt

**Correspondence:** Mostafa M. El-Gayar (mostafa_elgayar@mans.edu.eg)

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## ABSTRACT

The interest of security professionals and end users has made it a significant issue. Phishing is a deceptive practice in which a malicious individual pretends to be a trusted entity or organization in order to trick an unsuspecting victim into disclosing personal information online. The attacker creates an identical duplicate of the actual website. The address of the scamming web page is subsequently distributed to many web users through emails and various contact forms. Significant technological advancements in cybersecurity have occurred, especially in identifying phishing attacks automatically. However, much potential remains for improvement as hostile attackers create new technology. This paper systematically reviews studies on phishing detection between 2019 and 2025, with a particular focus on machine learning (ML) and deep learning (DL) approaches. A structured protocol based on PRISMA criteria and a ten-item quality assessment was applied. Selected studies were then compared in terms of datasets, characteristics, models, verification methods, and evaluation criteria. This review also compares and highlights similarities and differences in dataset recency, class imbalances, data leakage, time skew, and external verification, and offers suggestions for more effective and applicable phishing detection systems in other fields.

## 1 | Introduction

Although there are some similarities between cloud computing security and traditional on-premises network security, they differ fundamentally in their focus on protecting virtual environments. Like on-premises network security, cloud security aims to protect network resources from external threats. Cloud security solutions manage access requests and grant users the necessary permissions to access resources [1]. These solutions provide administrators with comprehensive security visibility through real-time alerts and audit logs, as well as the ability to protect against malware and unauthorized data transfers. However, unlike traditional security approaches, cloud-based protection systems operate externally and utilize software-defined network tools. Virtual gateways situated in the cloud ensure protection for applications and data regardless of user location [2].

Cloud attacks are defined as cyberattacks that target cloud service platforms, including computing and storage services, as well as applications hosted in a platform-as-a-service (PaaS) or software-as-a-service (SaaS) environment. These attacks can have serious consequences, such as data loss and leakage, unauthorized access to confidential information, and service disruptions [3]. SaaS-based phishing is a well-known threat,

Mostafa M. El-Gayar and Tarek M. Hassan contributed equally to this work.

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contributing to over 90% of data breaches, encompassing various tactics such as stolen credentials and malicious uniform resource locators (URLs). With the growing dependence on cloud-computing-based enterprises and individuals for data storage and processing, the number of potential targets for attackers is increasing proportionately. Many organizations may not be aware of the risks and vulnerabilities of cloud computing or may not be taking sufficient measures to mitigate these threats [4, 5].

The importance of cloud computing to businesses throughout the globe is increasing, which is causing rapid evolution in cloud security. Orca Security's report focuses on data breaches, system compromises, and supply chain attacks within cloud environments. The findings highlight that 45% of organizations experienced a data breach in the past year, and 38% reported system compromises. These statistics emphasize the need for robust cloud security measures and risk prioritization to mitigate these threats [6].

This research was conducted by cybersecurity insiders and Check Point Software Technologies to examine how businesses handle cloud security issues. According to the survey, to adequately manage security risks and safeguard their cloud assets, 72% of organizations are increasing their cloud security expenditures, and 65% are using new security solutions [7].

This significant increase underscores the need for organizations to adapt their core security measures to artificial intelligence (AI) and prioritize preventing evolving threats. The report also indicates that phishing remains the most common type of attack, accounting for 74% of all attacks experienced by victims [8]. A report from Palo Alto Networks analyzes cloud Security State, highlighting the balance between tools, vendors, and business requirements. The report revealed that 52% of organizations regularly use cloud security tools and 49% collaborate with vendors and stakeholders [9].

A Microsoft report discusses the risks and challenges of securing multicloud computing environments. This report emphasizes the need for comprehensive security strategies that can manage the complexities and vulnerabilities across multiple cloud computing platforms. According to the report, 68% of organizations use multicloud environments, and 42% of them have already experienced a security incident in these environments [10]. A Fortinet report provides key insights into how companies use cloud computing technologies and the security challenges they face. According to the report, 60% of companies have implemented cloud security training programs, while 58% use automated security solutions to strengthen their defenses against constantly evolving threats [11]. Phishing attacks remained high between 2023 and 2025, highlighting the continued growth of this threat. The highest quarterly figure was recorded in the second quarter of 2023, with 1,286,208 attacks. Despite fluctuations in subsequent quarters, phishing activity remained significant. In the second quarter of 2025, there were 1,130,393 attacks, before declining to 892,494 in the third quarter and then to 853,244 in the fourth quarter, a 4% decrease between the two-quarters. The Anti-Phishing Working Group (APWG) projected that the number of phishing attacks would reach 3.8 million by 2025 and 3.76 million by 2024, confirming the continued growth of this phenomenon. Simultaneously, spam attacks decreased by 45%, from 81,710 attacks in the third quarter of 2025 to 45,355 in the fourth quarter. This decline is primarily attributed to improved email filtering mechanisms that limit the spread of phishing links, as well as enhanced reporting channels. Despite these advancements in protection, phishing remains a widespread and sophisticated cyber threat, as shown in Figure 1 [12].

Phishing, a form of social engineering attack, is the primary technique used by cybercriminals to steal personal data from internet users without their permission [13]. These hackers use various methods to obtain sensitive information from unsuspecting users, such as credit card details, usernames, and passwords. In addition, they use phishing attacks to spread malware across networks [14].

Cyber fraud is diverse and represents a significant global threat. The latest Phishing Threat Trends Report (March 2025) revealed a significant increase in the scope and sophistication of these attacks. These attacks represented a 17.3% increase in phishing emails and a 53.5% year-over-year increase in phishing attacks. Furthermore, 82.6% of the attacks are AI-powered and sophisticated. In addition, attacks from compromised accounts increased by 57.9%, reflecting the increasing exploitation of trusted infrastructure by attackers. These findings highlight the ongoing changes and developments in attack methods, and organizations must be fully aware of and proactively prepared to address these challenges by enhancing their security measures, as shown in Figure 2 [15].

According to Hawks Hunt's "Phishing Trends 2025" report, phishing attacks are becoming more sophisticated, widespread, and dangerous. The report indicates that email phishing attacks target 64% of businesses. While approximately 40% of phishing attacks target email collaboration platforms, nearly 50% aim to steal user credentials, primarily targeting cloud services. This study analyzes data from over 2.5 million organizations, users,

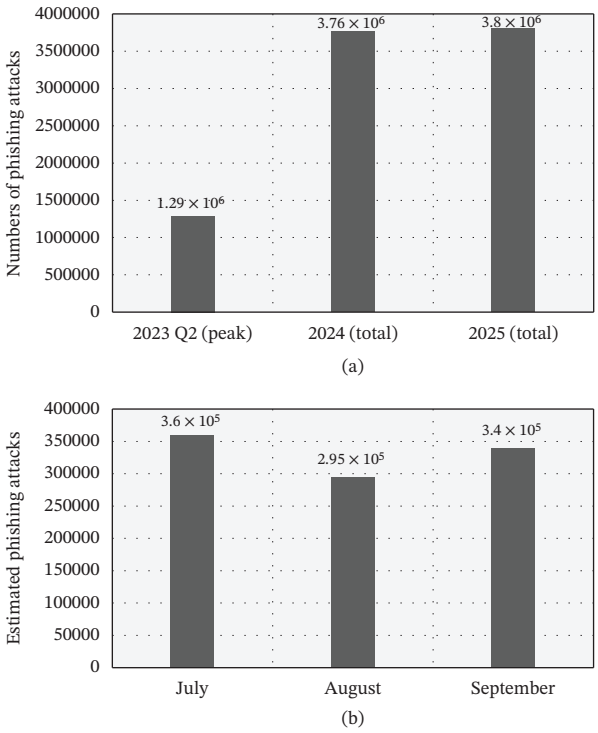



**FIGURE 1** | Phishing attack statistics: (a) year-on-year comparison and (b) monthly estimates for the third quarter of 2025 [12].



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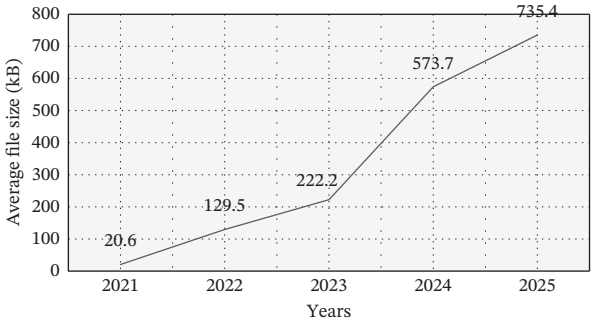



**FIGURE 2** | Average HTML file size observed in phishing emails (2021–2025) [15].

and 50 million phishing attacks [16]. The Zscaler ThreatLabz 2025 Phishing Report indicates a shift toward more targeted and sophisticated phishing attacks. Despite a 20% global decline in overall phishing volume, attackers increasingly focus on high-value departments such as IT, HR, and Finance. The report, based on over 2 billion blocked attempts, also reveals a 224% surge in attacks targeting the education sector and highlights the growing use of generative AI to create more evasive phishing messages. These results demonstrate strategic and technological developments in the field of phishing threats [17]. A 2025 report by Push Security revealed a phishing attack targeting LinkedIn users via direct messages rather than traditional email. Nonemail communication channels, such as social media and collaboration platforms, accounted for approximately 34% of detected phishing attacks. In some cases, the campaign used attack information management (AITM) techniques to steal login credentials and increase user engagement by exploiting trusted corporate communication channels. These findings suggest that multichannel phishing tactics are increasingly being used to bypass traditional email-based security measures [18]. According to the Abnormal Security report for the first half of 2024, QR code phishing attacks targeting executives are 42 times more likely to succeed. The report also focuses on the specific strategies used by attackers [19]. The University of California, San Francisco (UCSF), has provided updates to its Computer Science Department for May and June 2024, addressing recent phishing attacks that use the Carrier411 brand and compromised accounts to distribute malicious attachments. This demonstrates the variety of tactics employed by fraudsters [20]. Finally, data from the SSL Store show that Microsoft is the most imitated brand in phishing attacks; it was imitated in 43.1% of cases [21].

The number of malicious features allows for the detection of phishing websites, the identification of IP addresses within URLs, and the pinpointing of URL structures containing specific malicious characters and other indicators [4–6]. Inexperienced computer users can fall victim to the illusion of counterfeit web pages appearing legitimate. Several methods have been implemented to combat phishing and automatically detect phishing websites, including the creation of a main URL list called a blacklist-based detection method that categorizes websites in the same way as phishing sites [22]. When a user requests a web page, if its URL matches an entry on the blacklist, the connection to the requested website is promptly blocked. Phishing attacks can take many different forms. Some of the more commonly observed are whaling, malware-based phishing, spear phishing, angler phishing, email phishing, website spoofing, and smishing and vishing, with the following percentages, respectively: 15%, 20%, 25%, 10%, 30%, and 10%, and phishing is a favored technique among fraudsters because of its effectiveness in deceiving users into divulging sensitive information or accessing critical accounts and services, resulting in financial loss and identity theft. This method exploits the ease with which individuals are deceived into clicking malicious links, which can compromise a computer's security measures or allow perpetrators to circumvent modern authentication systems, as shown in Figure 3 [24]. By defending against these attacks, individuals can protect their personal and financial data from being misused. Preventing phishing attacks helps reduce the risk of identity theft, financial fraud, and unauthorized access to sensitive accounts, as shown in Figure 4 [25].

Intrusion detection systems (IDSs) are vital for safeguarding computer systems and networks through the detection of illegal or suspicious activity and the notification of administrators of possible dangers. IDSs can be in a variety of forms, such as host-based intrusion detection systems (HIDSs) that monitor specific devices and network-based intrusion detection systems (NIDSs) that monitor network traffic [26]. While an anomaly-based IDS looks for deviations from normal activity, a signature-based IDS uses known attack signatures to identify threats. Both techniques are combined by a hybrid IDS for thorough detection. While application-based IDSs keep an eye on particular software applications, protocol-based IDSs concentrate on particular communication protocols. Combining the advantages of each type can result in strong security against a variety of cyberthreats [27].

It is crucial to implement a comprehensive security program and develop it using diverse methods and advanced security tools, ranging from encryption protocols to advanced AI solutions, to counter sophisticated attacks and protect cloud computing systems [28]. Cryptography, which offers tools for data encryption, communication security, and user authentication, is the foundation of cloud security. These features help combat future phishing attacks. Furthermore, by contributing to early threat detection, anomaly identification, and adaptive response mechanisms, AI technologies greatly enhance traditional security measures and strengthen the ability of cloud infrastructure to counter phishing attacks, as shown in Figures 5 and 6 [12].

Despite the availability of various technologies and tools for detecting phishing attacks, the risk of falling victim remains high. These attacks have become increasingly sophisticated and

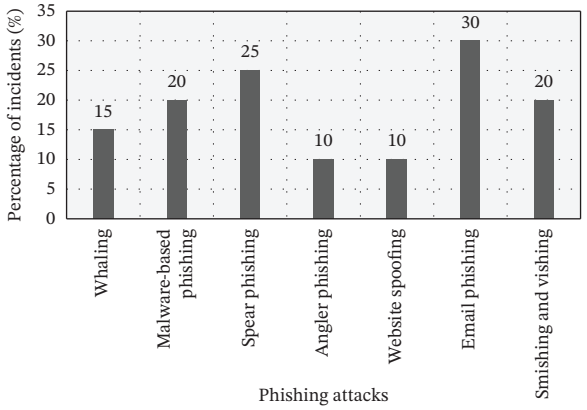



**FIGURE 3** | Phishing attack types [23].



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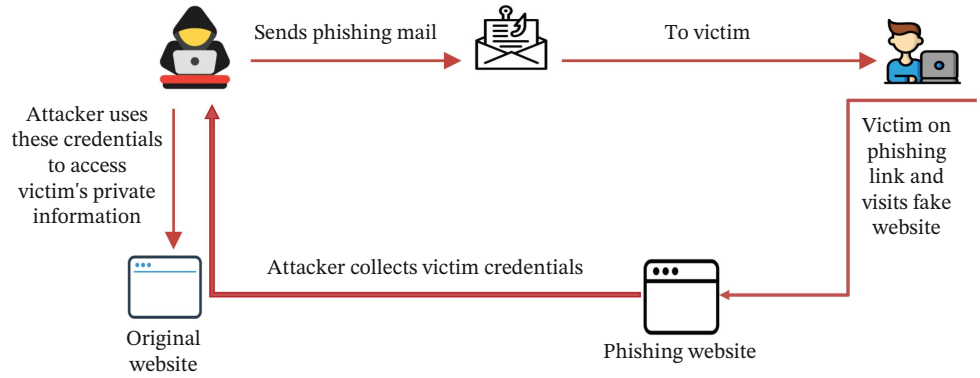



**FIGURE 4** | Phishing attack diagram.

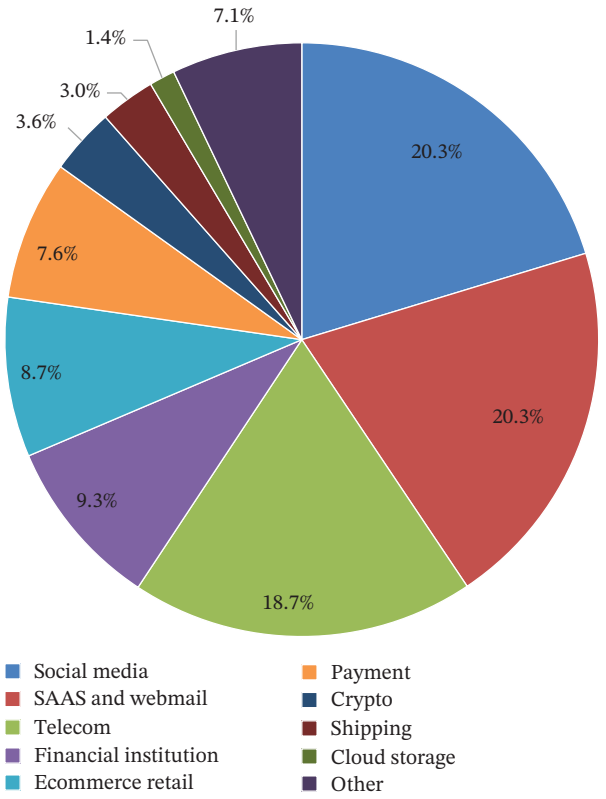



**FIGURE 5** | Most-targeted industry sectors, Q4 2025 [12].

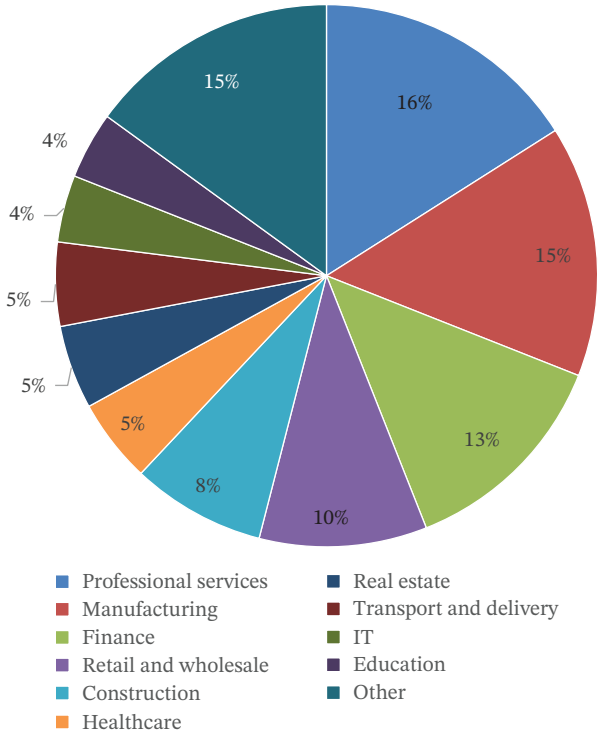



**FIGURE 6** | Industries most targeted by QR code phishing, Q4 2025 [12].

difficult to detect. This increase is due to the rise in cybercrime on email networks, where traditional detection methods are often ineffective [29].

It also drastically reduces the effectiveness and productivity of online users and companies unless an intelligent anti-phishing solution is developed. Effective phishing website efforts can lead to major loss of data, compromised login credentials, ransomware infestation, and loss of money. Therefore, there is a requirement for a phishing detection system that can accurately and effectively detect phishing attacks in real time, protect users from these scams, and reduce their impact on individuals, companies, and institutions [14].

Preventing phishing attacks is essential for protecting personal and financial information, maintaining online privacy, reducing financial risks, safeguarding company stock values, preserving trust and reputation, increasing cybersecurity awareness, and ultimately improving overall security. Through effective and continuous defense against phishing attacks and proactive security measures, individuals and organizations can create safer environments.

Several recent surveys have analyzed phishing detection techniques using machine learning (ML) and deep learning (DL); however, numerous gaps remain that need to be addressed. Most current reviews tend to focus on classifying detections or providing an overview of reported performance without adequately considering the quality of the data used and the evaluation methods employed. In particular, not all previous surveys address issues such as the novelty of datasets, class imbalances, sample duplication, data leakage, time lag, external validation, and real-world generalizability.



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Furthermore, direct comparisons between reported results are not always possible because of variations in datasets, dataset representations, validation protocols, and evaluation criteria. Therefore, a systematic review is needed that not only summarizes the latest anti-phishing techniques but also critically examines methodological quality, dataset characteristics, validation methods, and the usability of reported evidence.

The motivation for this study was the need for a formal and current review of phishing detection research. Although many reviews have been published on phishing detection methods, some of the existing surveys tend to limit their scope to specific methods, datasets, or evaluation contexts. Furthermore, the topic of data quality, class imbalance, duplicate data, data leakage, temporal drift, external validation, and real-world generalizability is not always addressed. Thus, the present systematic literature review gives a thorough examination of the latest phishing detection research published in the timeframe of 2019–2025. It examines the detection methods used, types of data, representations of features, validation procedures, evaluation measures, methodological quality, limitations, and practicality. Therefore, the significant contributions of this study are summarized as follows:

- A systematic review of recent phishing detection studies is presented, including ML, DL, hybrid, URL-based, content-based, visual, and multimodal approaches.
- A systematic survey of widely used datasets, data sources, feature representations, validation protocols, evaluation metrics, and reported performances.
- A methodological quality assessment of the 32 studies that were included in the study, focusing on the description of the dataset, distribution of classes, design of external or temporal evaluation, presence of data leakage, and validation design.
- Discussion of representative examples of ML and DL studies and their respective advantages, disadvantages, and applications.
- Recognition of recurring research issues, such as aging datasets, class imbalance, duplicate samples, limited external validation, temporal drift, real-time limitations, and false-positive and false-negative errors.
- Suggestions for future research include creating better, more generalizable, leakage-proof, and real-time phishing detection systems.

This paper is structured into four sections, as shown in Figure 7.

- Section 2 outlines the materials and methods for this systematic review, such as the review scope, research questions, information sources, search, inclusion and exclusion criteria, study selection and PRISMA flow, quality assessment, and data extraction and synthesis procedures.
- Section 3 presents the findings and discussion, including background on phishing detection techniques, analysis of commonly used datasets, a review of selected studies based on ML and DL, key challenges, and future research directions.
- Section 4 concludes the paper by summarizing the main findings and implications of the review.

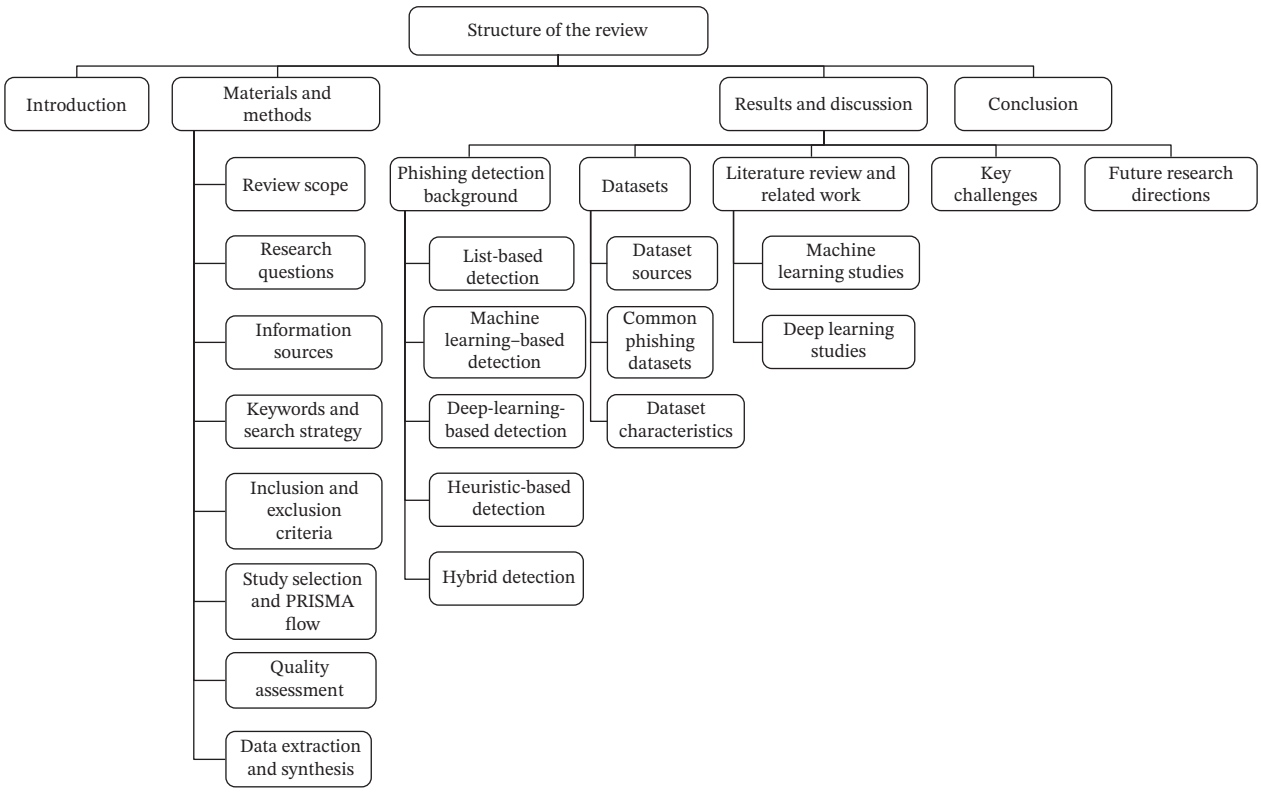



**FIGURE 7** | The structure of the review.



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## 2 | Materials and Methods

### 2.1 | Review Scope

This systematic literature review examines recent (2019–2025) intelligent methods for detecting phishing. The review highlights studies published in peer-reviewed journals that suggest, evaluate, or comparatively analyze ML, DL, hybrid, visual, content-based, URL-based, email-based, or other AI-supported techniques for phishing detection. Phishing detection has been studied from an ML perspective, an DL perspective, and an intelligent phishing detection perspective in recent reviews and systematic literature reviews. In accordance with these recent review efforts [13, 30–32], the present study follows the structured systematic review process of the PRISMA guideline, quality assessment of the included studies, and structured evidence synthesis.

This review considered not only the reported performance of phishing detection methods but also the datasets, modalities, feature representations, evaluation protocols, and methodological limitations that could affect the reported results. Special emphasis was placed on dataset freshness, class imbalance, duplicated samples, possible data leakage, temporal drift, external validation, real-world applicability, and the capability of the proposed methods to generalize to new phishing campaigns.

### 2.2 | Research Questions

- RQ1. What phishing detection methods have been suggested between 2019 and 2025?
- RQ2. What are the most frequent datasets, features, and algorithms?
- RQ3. What are the differences between the ML, DL, and hybrid methods?
- RQ4. What are some problems with datasets that can impact the reliability of the results?
- RQ5. What studies did you assess for generalization, real-time detection, or robustness?
- RQ6. What are the most important restrictions, gaps in knowledge, and future directions?

### 2.3 | Information Sources

Structured searches of IEEE Xplore, SpringerLink, ScienceDirect, MDPI, and The Science and Information Organization (SIORG) were conducted to identify relevant studies. Each source had its own search syntax, and a common Boolean search strategy was used. Only English-language publications between 2019 and 2025 were used for searching. The first results of the search were then filtered based on the following inclusion and exclusion criteria. Table 1 outlines the search terms, search dates, filters, and number of hits found in each source.

### 2.4 | Keywords

Initially, we employed targeted keywords related to phishing attack detection, such as “phishing detection techniques” and “phishing prevention methods,” in our search process. Following this initial search, additional keywords were gathered from the articles retrieved to expand the scope of our search. In addition, we have included other terms such as “Machine learning for phishing detection,” “Deep learning for phishing attack detection,” “Email security solutions,” and “phishing website detection” to improve our understanding of the research area. The keywords were combined using the Boolean operators “AND” and “OR” to retrieve studies related to phishing detection and AI.

### 2.5 | Inclusion and Exclusion of Article Criteria

To identify the publications most relevant to our study, we used inclusion and exclusion criteria specifically designed for our research objectives. Studies that met the inclusion criteria were considered relevant, while those that did not were excluded, as shown in Figure 8.

### 2.6 | Selection of the Articles and PRISMA Flow

The study selection process consisted of three steps. First, 170 references were retrieved from the IEEE Xplore, SpringerLink, The SIORG, ScienceDirect, and MDPI databases. After removing 28 duplicate references, 142 studies remained for review of their titles and abstracts. In this step, 84 references were excluded for not being relevant to the journal’s scope or for not meeting the predefined inclusion criteria. The full texts of the remaining 58 articles were reviewed for relevance. Following this review, 26 articles were excluded, leaving 32 studies for the final qualitative analysis and comparisons. The PRISMA [33] diagram illustrates the entire process: identification, selection, relevance review, and inclusion, as shown in Figure 9.

### 2.7 | Quality Assessment

The qualitative study methodology includes a structured evaluation approach, encompassing research description, data description, preprocessing, characteristic representation, model construction, validation, evaluation criteria, and more, as shown in Table 2. Evaluation thresholds were set at 1, 0.5, and 0 points, indicating partial or complete failure to meet the established criteria. The total score was then used to classify the studies into three categories: high quality, medium quality, and low quality [31].

Quality assessment showed that none of the included studies was classified as low quality. In contrast, 11 and 21 studies were classified as moderate and high quality, respectively (scores of 5.0–7.5), and 21 were classified as high quality (scores of 8.0–9.5, respectively). As shown in Figure 10, most of the selected studies showed generally good methodological quality, as evidenced by the peak number of studies at 8.0 and 9.0. Overall, the results indicate that most studies clearly stated their objectives, datasets, models, and evaluation metrics. Simultaneously, some topics, such as data leakage analysis and external and temporal validation, were not consistently addressed.

### 2.8 | Data Extraction and Synthesis

A structured data extraction model was used to collect relevant information from individual studies. The extracted data included publication year, phishing type, detection method, database name and source, category size and distribution, feature representation, preprocessing methods, ML or DL, validation protocol, assessment measures, reported effects, strengths and



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**TABLE 1** | Search strategy used to identify relevant studies.

| No. | Source | Search string | Search date | Applied filters | Records |
|---|---|---|---|---|---|
| 1 | IEEE Xplore | (“phishing detection” OR “phishing website detection” OR “phishing URL detection” OR “phishing email detection”) AND (“machine learning” OR “deep learning” OR “artificial intelligence” OR “hybrid detection”) | May 2025 | Years: 2019–2025; English; Journal articles | 42 |
| 2 | SpringerLink | (“phishing detection” OR “phishing website detection” OR “phishing URL detection” OR “phishing email detection”) AND (“machine learning” OR “deep learning” OR “artificial intelligence” OR “hybrid detection”) | February 2025 | Years: 2019–2025; English; Journal articles | 40 |
| 3 | The Science and Information Organization | (“phishing detection” OR “phishing website detection” OR “phishing URL detection”) AND (“machine learning” OR “deep learning” OR “artificial intelligence”) | March 2025 | Years: 2019–2025; English; Journal articles | 36 |
| 4 | ScienceDirect | (“phishing detection” OR “phishing website detection” OR “phishing URL detection” OR “phishing email detection”) AND (“machine learning” OR “deep learning” OR “artificial intelligence” OR “hybrid detection”) | July 2025 | Years: 2019–2025; English; Research articles | 32 |
| 5 | MDPI | (“phishing detection” OR “phishing website detection” OR “phishing URL detection” OR “phishing email detection”) AND (“machine learning” OR “deep learning” OR “artificial intelligence” OR “hybrid detection”) | September 2025 | Years: 2019–2025; English; Journal articles | 20 |
| | Total | | | | 170 |



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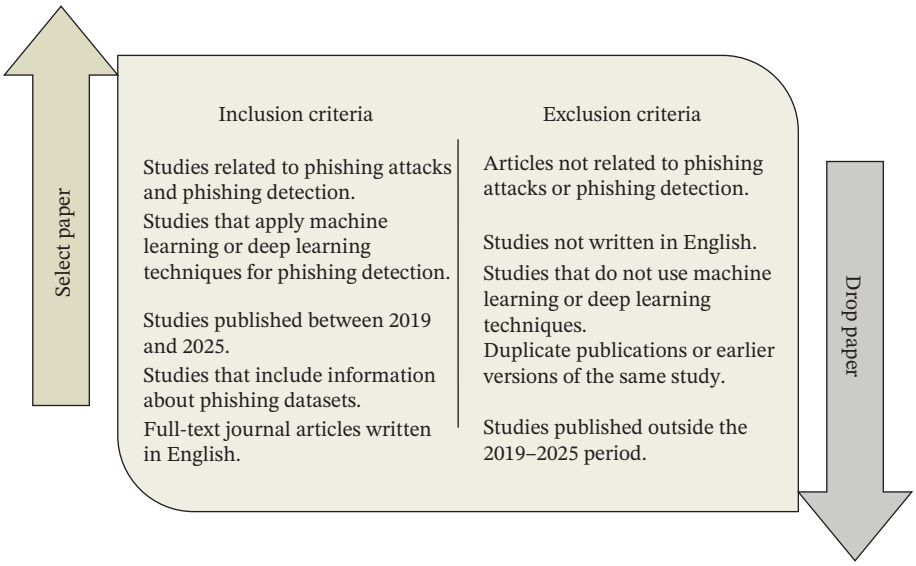



**FIGURE 8** | The specific criteria utilized for inclusion and exclusion in our study techniques.

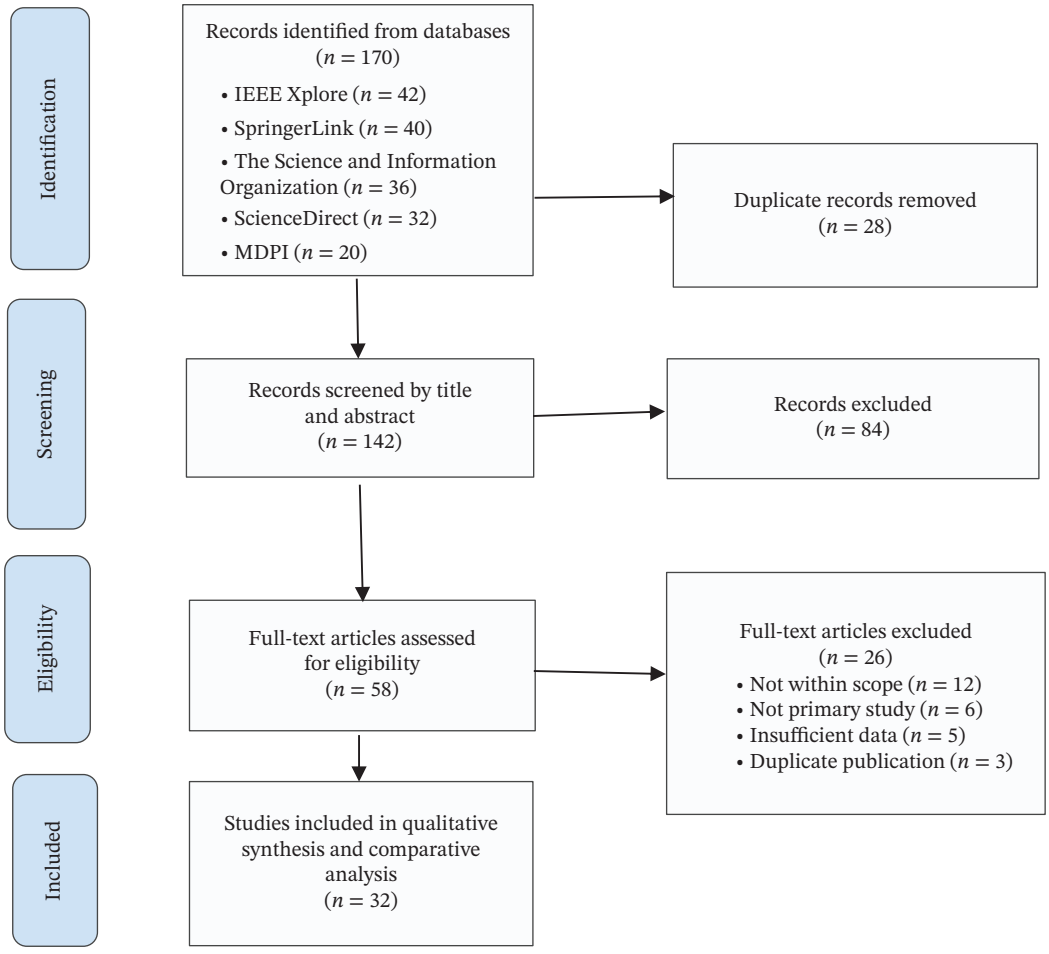



**FIGURE 9** | PRISMA flow diagram of the study selection process.

weaknesses, and recommendations for future studies. This information was organized into comparison tables and subject categories. Due to the diversity of databases, category distributions, validation strategies, and impact measures used in the analyzed studies, a formal meta-analysis was deemed inappropriate. Therefore, a narrative and descriptive synthesis was conducted. The studies were compared based on detection methods, databases, feature types, validation methods,



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**TABLE 2** | Quality-assessment criteria used for the included studies.

| ID | Quality-assessment criterion |
|---|---|
| QA1 | Is the research objective clearly stated? |
| QA2 | Is the dataset source and size clearly reported? |
| QA3 | Is the class distribution described? |
| QA4 | Are preprocessing and feature-representation procedures explained? |
| QA5 | Is the model architecture or classifier configuration clearly described? |
| QA6 | Is the validation protocol clearly reported? |
| QA7 | Are appropriate evaluation metrics presented? |
| QA8 | Are dataset imbalance, duplication, or data-leakage risks considered? |
| QA9 | Is external, cross-dataset, or temporal validation performed? |
| QA10 | Are limitations and future directions discussed? |

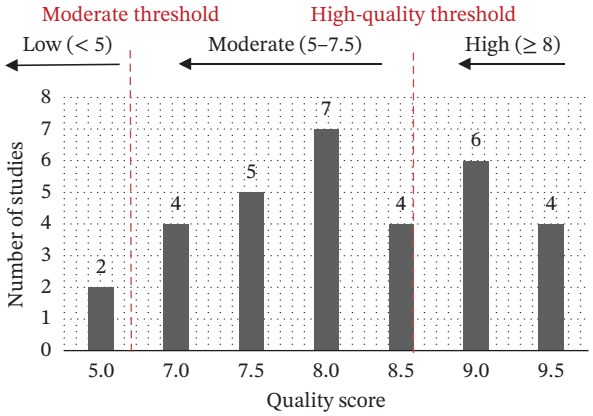



**FIGURE 10** | Distribution of the total methodological quality scores assigned to the 32 included studies.

methodological quality, and reported outcomes. Regression analyses were also used to group the most common data collection methods and assessment measures in the selected studies.

## 3 | Results and Discussion

### 3.1 | Background on Phishing Detection Techniques

In recent years, numerous and increasingly diverse technologies have been developed to detect phishing attacks, including ML and DL. These technologies use algorithms and models to analyze the features of various traditional phishing techniques. Below, we provide a comprehensive overview and details on the available options, illustrating their potential for phishing detection. As shown in Figure 11, it is classified into five groups in this study.

#### 3.1.1 | List-Based Phishing Detection Techniques

List-based phishing detection depends on whitelists and blacklists created from reports from the community. URLs that the user has trusted are in the whitelist. Conversely, a URL is regarded as malicious when it is included in the blacklist [34]. List-based phishing detection techniques are generally classified into two primary categories: blacklist-based and whitelist-based approaches. Blacklist-based approaches compare the suspicious domain with a predefined list of known phishing domains. Although effective, this method is limited because newly launched phishing websites may not be immediately added to the blacklist. Whitelist-based approaches analyze the features of a website against common features found in phishing sites, which allows for the detection of zero-day attacks [35]. However, these methods can produce a high false-positive rate, making it difficult to detect embedded elements within web pages. Although these methods can detect embedded elements, they are less effective against zero-day attacks. Each technology has its advantages and disadvantages, and no single technology can comprehensively detect all types of phishing attacks [36].

#### 3.1.2 | ML Detection Techniques

ML techniques used to detect phishing attacks have undergone significant development to counter the ever-evolving methods used by cybercriminals and to keep pace with current advancements. These techniques fall into two categories: supervised learning and unsupervised learning. First, supervised learning algorithms, such as support vector machines (SVMs), logistic regression (LR), decision trees (DTs), random forests (RFs), and K-nearest neighbors (KNN), are trained on datasets containing both legitimate and deceptive URLs or email addresses. These models learn to differentiate between phishing attacks and legitimate attacks based on features such as URL elements, email addresses, and text content, as shown in Figure 12 [37]. Unsupervised learning techniques, such as clustering and anomaly detection, identify phishing attacks by recognizing deviations from normal patterns of legitimate activity without the need for prior classification, as confirmed by a comprehensive study [38].

These techniques utilize various properties, such as URL information, website structure, and JavaScript functionality, to create accurate representations of phishing URLs and their associated websites, such as URL information, web page structure, and JavaScript functionality to generate accurate representations of phishing URLs and their associated pages. This systematic approach allows for the collection of large amounts of phishing data, which is essential for building a reliable model. ML classifiers are then trained on these datasets to detect delicate patterns indicative of phishing attacks, as described in this research paper [39]. Especially, this method has proven highly effective in



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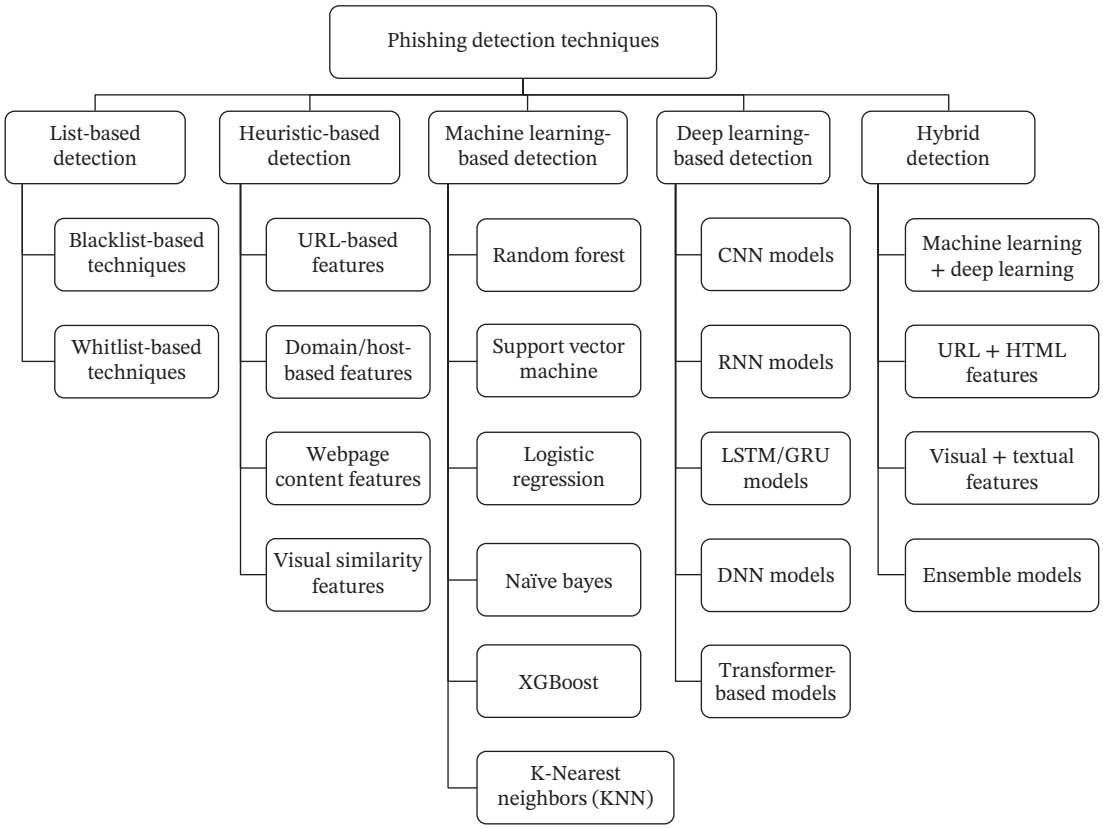



**FIGURE 11** | The structure of phishing website detection techniques.

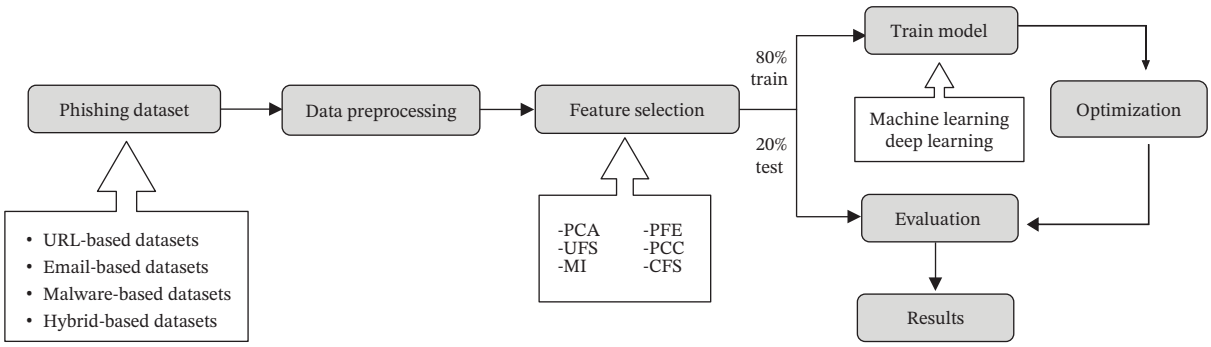



**FIGURE 12** | General pipeline for phishing detection systems.

the field of big data, which is characterized by its speed, diversity, volume, value, and reliability. One of the most significant advantages of ML-based classifiers is their high accuracy, exceeding 99%. As studies show, this makes them a leading solution for detecting phishing sites [38].

### 3.1.3 | DL Detection Techniques

DL algorithms have recently made significant strides in numerous research areas, including image classification and facial recognition. They have also been successfully applied to address several types of cybersecurity challenges, such as malware detection, phishing, spam, and protection against hacked websites [40]. In recent years, a wide range of DL algorithms have been employed in diverse operations to address cybersecurity risks and challenges. These algorithms include deep neural networks (DNNs), recurrent neural networks (RNNs), feedforward neural networks (FNNs), convolutional neural networks (CNNs), deep belief networks (DBNs), and deep autoencoders, as demonstrated in a comprehensive study [41].

A DNN, characterized by multiple hidden layers, is a sophisticated type of ANN used for classification and regression. It typically includes an input layer, an output layer, and hidden layers, and is suitable for solving a wide variety of problems [42]. Unlike FNNs, RNNs use the outputs of previous stages as current inputs. This makes them well-suited for sequential tasks, such as predicting the next word. These networks process input data repeatedly using a hidden state that stores the sequential data and allows the network to remember previous information [43]. These highly advanced DL techniques have shown considerable potential for improving the accuracy and efficiency of phishing



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website detection and represent a significant advance in cybersecurity research and application [44]. Transformer-based models are DL architectures that are capable of learning long-range relationships in input data via self-attention. Their capacity to learn such complex representations with minimal manual feature engineering has made them popular for use in cybersecurity-related applications [45].

### 3.1.4 | Heuristics-Based Detection Techniques

In heuristic-based detection, a set of predetermined rules and manually picked features are employed to identify phishing websites based on the content of a web page, the domain information, or the URL. The techniques identify suspicious characteristics that could potentially suggest malicious behavior and can be applied to detect attacks that have not yet been seen or zero-day attacks [46]. However, they are only effective when the rules and features chosen are good, and when the legitimate and phishing websites have similar characteristics, false-positive or false-negative predictions can be made. One of the challenges in heuristic-based detection is concept drift, which means that the phishing technique and the web page structure evolve. The heuristic-based approaches can be broadly divided into three categories: URL- and host-based approaches, content-based, and visual similarity-based approaches [36, 47].

URL- and Host-Based Approaches: These are based on features that are not central to the web page content. This can include the presence of SSL certificates, WHOIS data, domain age, URL structure, and more. Often, heuristic rules are used to identify anomalous URLs, such as those that contain unusually long URLs, use an IP address in the URL, have too many subdomains, contain nonstandard symbols, or have recently registered domains [48].

Content-Based Approaches: These methods examine the text, graphics, and HTML tags that make up a web page. The number of links, the existence of specific keywords, and the web page's structure are examples of features that may be extracted. Here, heuristic approaches might search for discrepancies between links that appear to be displayed and those that do not, login forms on insecure websites, and other indicators of phishing attacks within the text of the page itself [49].

Visually Similarity-Based Approaches: These methods compare the visual appearance of a suspicious phishing page with that of legitimate websites. They use elements such as layout, visual identity, and overall design to identify pages that visually resemble popular websites. This method frequently uses ML and image processing tools to find commonalities that text analysis alone could miss [50].

### 3.1.5 | Hybrid Detection Techniques

Hybrid phishing detection techniques can take advantage of complementary sources of phishing evidence by using multiple feature types, learning paradigms, or model architectures. They can introduce ML and DL, or use a combination of URL and HTML or web page-content features, or a fusion of visual and textual information, or ensemble models. For instance, [51] extracted the URL, textual, and frame information by using long short-term memory (LSTM) and web page images by using CNN in combination for phishing detection to create a multimodal phishing detection system. Reference [52] fused visual and textual identity information for better detection of deceptive web pages, and [53] used a fusion of DNN and LSTM architecture for phishing URL detection. Other studies fused handcrafted and automatically learned features: [54] fused optimized artificial features with CNN-based deep URL representations and an RF classifier. Reference [55] also suggested a hybrid approach by combining ResNeXt, GRU, attention, autoencoder, and ResNet representations. Such methods can offer more comprehensive and reliable phishing evidence than single-feature and/or single-model methods. They often, however, need more computational power, more complex training algorithms, and careful validation to prevent overfitting and data leakage.

## 3.2 | Datasets

Large training datasets are essential for developing effective models. With the increase in phishing attacks, the requirement for new, high-quality datasets has increased, as improving model performance is crucial. Therefore, it is important to examine existing datasets to better understand this phenomenon. Tables 3 and 4 present the common and secure datasets and phishing websites. This section provides an overview of the most frequently used datasets for phishing detection and the most cited sources in the current research.

### 3.2.1 | Phishing_Legitimate_Full.csv

This dataset, which was made available by Chiew et al. [65], includes 48 features that were taken from 5000 legitimate and 5000 phishing websites. While the authentic websites were compiled via Alexa and Common Crawl, PhishTank and Open Phish provided the list of phishing websites. The phishing and legitimate samples were collected during separate periods between January 2015 and June 2017.

### 3.2.2 | Dataset_Full.csv

This dataset [66] was split into two dataset versions: 58,645 small and 88, 647 full versions that were classified as phishing and legitimate, respectively. In all, there are 58,645 cases in the limited dataset, consisting of 30,647 phishing websites (labeled as 1) and 27,998 authentic websites (labeled as 0). The complete dataset consists of 88,647 cases, of which 30,647 are classified as phishing and 58,000 as legitimate. 111 features define each instance.

**TABLE 3** | List of the dataset resources.

| Reference | Source |
|---|---|
| [56] | PhishTank |
| [57] | Phishing Army |
| [58] | Kaggle |
| [59] | JoeWein |
| [60] | Phishing Dataset |
| [61] | Common Crawl |
| [62] | Yandex |
| [63] | OpenPhish |
| [64] | Phishload |



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**TABLE 4** | Critical comparison of common phishing detection datasets.

| Dataset | Collection year/period | Dataset size and class distribution | Phishing data source | Legitimate data source | Feature types/ modalities | Public availability | Imbalance ratio* | Main limitations | Suitability for modern phishing evaluation |
|---|---|---|---|---|---|---|---|---|---|
| Phishing Legitimate full.csv [65] | Samples collected during January–May 2015 and May–June 2017; dataset released in 2018 | 10,000 samples: 5000 phishing and 5000 legitimate web pages | PhishTank and OpenPhish | Alexa and Common Crawl | 48 URL-based and web page-based features | Publicly available | 1.00:1 | The samples are relatively old and may not represent recent phishing campaigns. The balanced class distribution may also differ from operational environments. The dataset depends mainly on predefined handcrafted features. | Suitable for controlled comparisons of traditional ML methods, but temporal and external validation using recent samples is required. |
| Dataset full.csv [66] | Released in 2020; the exact collection period is not explicitly reported | Limited version: 58,645 samples, including 30,647 phishing and 27,998 legitimate websites. Full version: 88,647 samples, including 30,647 phishing and 58,000 legitimate websites | Not explicitly reported in the available dataset documentation | Not explicitly reported in the available dataset documentation | 111 URL, domain, DNS, server, and web page-related features | Publicly available through Mendeley Data | 0.91:1 for the limited version; 1.89:1 for the full version | The sources and collection dates are insufficiently documented. Several features depend on domain, DNS, or external services, which may introduce missing values and increase real-time detection latency. | Useful for evaluating feature-based ML models, but source transparency, temporal freshness, and cross-dataset testing should be improved. |

(Continues)



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**TABLE 4** | (Continued)

| Dataset | Collection year/period | Dataset size and class distribution | Phishing data source | Legitimate data source | Feature types/ modalities | Public availability | Imbalance ratio* | Main limitations | Suitability for modern phishing evaluation |
|---|---|---|---|---|---|---|---|---|---|
| PILU-60K [67] | Phishing URLs collected from November 2019 to January 2020 | 60,000 URLs: 20,000 phishing, 20,000 legitimate index URLs, and 20,000 legitimate login URLs | PhishTank | Highly ranked legitimate websites; login pages were identified from legitimate domains | Raw URL strings | Publicly available | 2.00:1 | The dataset contains only URLs and excludes HTML, web page images, scripts, and domain-context information. Its samples are now temporally limited for evaluating recent campaigns. | Highly suitable for realistic URL-based evaluation because it distinguishes legitimate login pages from ordinary legitimate homepages. Temporal testing with newer URLs is still needed. |
| Dataset phishing.csv [68] | Released in 2021; the exact collection period is not explicitly reported | 11,430 samples: 50% phishing and 50% legitimate | Not explicitly reported in the available documentation | Not explicitly reported in the available documentation | 87 features: 56 URL structure and syntax features, 24 web page-content features, and 7 external-service features | Publicly available through Mendeley Data | 1.00:1 | The dataset is relatively small and artificially balanced. External-service features may be unavailable or slow during operational detection. The collection sources and dates are not clearly documented. | Suitable for comparing URL-, content-, and hybrid feature-based models, but less suitable for evaluating temporal robustness and large-scale real-world deployment. |

(Continues)



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**TABLE 4** | (Continued)

| Dataset | Collection year/period | Dataset size and class distribution | Phishing data source | Legitimate data source | Feature types/ modalities | Public availability | Imbalance ratio* | Main limitations | Suitability for modern phishing evaluation |
|---|---|---|---|---|---|---|---|---|---|
| PILU-90K [69] | Released in 2021; constructed to support comparison across different URL collection periods | 90,000 URLs: 30,000 phishing, 30,000 legitimate index URLs, and 30,000 legitimate login URLs | PhishTank | Legitimate index pages and automatically identified login pages from ranked legitimate domains | Raw URL strings | Publicly available | 2.00:1 | The dataset is restricted to URL strings and does not include web page content or visual evidence. Model performance may decrease when older training samples are evaluated against newer URLs. | Highly suitable for analyzing realistic login-page confusion and temporal performance degradation, but multimodal evaluation requires additional datasets. |
| Phishing Websites Dataset [70] | Released in 2021; the exact collection period is not clearly reported | 80,000 samples after preprocessing: 30,000 phishing and 50,000 legitimate websites | PhishTank, OpenPhish, and PhishRepo | Google search and the Ebbu 2017 dataset, as reported in the reviewed source | URL-based features | Publicly available | 1.67:1 | The dataset is moderately imbalanced. The distribution of samples across the individual sources, collection dates, duplicate-removal process, and leakage checks is not sufficiently documented. | Suitable for relatively large-scale URL classification, provided that deduplication, source-aware partitioning, and leakage analysis are conducted before evaluation. |

(Continues)



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**TABLE 4** | (Continued)

| Dataset | Collection year/period | Dataset size and class distribution | Phishing data source | Legitimate data source | Feature types/ modalities | Public availability | Imbalance ratio* | Main limitations | Suitability for modern phishing evaluation |
|---|---|---|---|---|---|---|---|---|---|
| PILWD-134K [71] | Collected from August 2019 to September 2020; released in 2022 | 134,000 balanced phishing and legitimate login-page samples | Phishing web pages collected for the PILWD dataset; the exact provider is not explicitly stated in the reviewed manuscript | Legitimate login web pages | URLs, HTML source code, web page screenshots, raw web page files, web technologies, and metadata | Publicly available | 1.00:1 | The collection period may not reflect current phishing techniques. Processing screenshots, HTML, and raw web page files increases storage requirements, computational cost, and detection latency. | Highly suitable for multimodal and hybrid evaluation because it supports URL-, content-, visual-, and metadata-based detection. Newer temporal samples remain necessary. |
| PhiUSIIL Phishing URL Dataset [72] | Released in 2024; described as containing relatively recent URLs, but the exact collection dates are not clearly stated | 235,795 samples: 100,945 phishing and 134,850 legitimate URLs | Not explicitly stated in the reviewed manuscript | Not explicitly stated in the reviewed manuscript | 54 URL-based and web page source-code features | Publicly available through the UCI ML Repository | 1.34:1 | The dataset has moderate class imbalance and relies largely on engineered URL and source-code features. Exact collection dates, source-specific distributions, and temporal independence require clearer reporting. | One of the largest and most recent datasets included in the review. It is suitable for modern ML evaluation, although independent temporal, cross-source, and leakage-aware testing is recommended. |

*Imbalance ratio = number of legitimate samples : number of phishing samples. A ratio of 1.00:1 represents a balanced dataset.



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### 3.2.3 | Dataset_Phishing.csv

This dataset, which was given by [68], contains 11,430 URLs with 87 extracted features. The dataset includes three distinct feature sets: 56 features from the URL structure, 24 features from the URL content, and 7 features from requests to external services. Fifty percent of the links in the dataset are phishing URLs, while the remaining 50% are legitimate URLs.

### 3.2.4 | Phishing Websites Dataset

This dataset, provided by [70], contains 80,000 instances (83,275 initially, with some removed during preprocessing), including 50,000 legitimate websites (labeled as 0) and 30,000 phishing websites (labeled as 1). These data were obtained from two sources: Google search and the Ebbu 2017 phishing dataset. The 30,000 phishing data points were gathered from three sources: PhishTank, OpenPhish, and PhishRepo.

### 3.2.5 | PhiUSIIL Phishing URL (Website)

This dataset [72] is a massive dataset containing 100,945 phishing URLs and 134,850 legitimate URLs. Most of the URLs examined when the dataset was created were among the most recent. The web pages and URLs' source code are used to extract features. CharContinuationRate, URLTitleMatchScore, URLCharProb, and TLDLegitimateProb are examples of features developed from preexisting features.

### 3.2.6 | Phishing Index Login URL Dataset (PILU-60K)

This dataset was provided by [67] created by the Vision and Intelligence System Research Group at the University of Léon. It consists of 60,000 URLs, which are equally distributed among three classes of URLs: 20,000 legitimate index URLs, 20,000 legitimate login URLs, and 20,000 phishing URLs. The dataset was created to assess the accuracy of the classification of ML models when differentiating between legitimate homepage and login URLs and phishing web pages. Phishing URLs were taken from Phish Tank from November 2019 to January 2020.

### 3.2.7 | Phishing Index Login URL (PILU-90K)

This dataset provided by [69] contains 90,000 URLs, including 60,000 legitimate URLs, including indexing and login websites, and 30,000 phishing URLs. It was used to illustrate the decline in model accuracy over time when trained on older datasets and tested on newer URLs.

### 3.2.8 | Phishing Index Login Website Dataset (PILWD)

This dataset was provided by [71] created by the Vision and Intelligence System Research Group from the University of Léon; it comprises 134,000 samples of authentic and balanced logon resources from phishing and legitimate websites. Sampling was performed from August 2019 to September 2020. In addition to the URLs, the dataset contains HTML source code, screenshots of the web pages, raw files on the web pages, web-technology information, and metadata, making it appropriate for testing multimodal phishing detection methods.

Different types of data can be employed to detect phishing attacks, including URL-based [40, 73], content-based [74], and hybrid-based methods [75, 76]. URL-based methods analyze URL information exclusively, without considering additional web page details such as content or title. The advantage of this method is that it allows users to detect phishing attacks without clicking on a URL, thus reducing the risk of malware being downloaded and installed [77]. However, relying solely on data derived from URL-based features ignores important aspects of the phishing site, such as the page title and source code, and it is particularly difficult to analyze shortened URLs using only this method. Content-based methods, on the other hand, extract information from website content, including images, JavaScript, text, and HTML code [78]. While this method is effective, it requires users or systems to access the website to extract the content, thereby increasing the risk of accidentally downloading and installing malware. Therefore, in hybrid approaches, we combine the advantages of both URL-based and content-based methods to maximize their benefits [79].

URL-based phishing detection methods analyze the features of URLs to identify potential attacks. These methods typically use ML algorithms to classify website addresses as malicious or legitimate based on criteria such as domain name, domain age, address length, presence of special characters, and use of IP addresses instead of domain names. Research in this area focuses on developing ML models, such as SVM and RF, to accurately detect phishing website URLs [80].

Content-based methods focus on analyzing the content of web pages, such as HTML structure, embedded scripts, and visual scripts to detect phishing attacks [81]. These methods look for features such as forms requesting personal data, keywords related to sensitive information, and the overall structure of websites. Although it often requires more computing resources than methods based on website addresses, it offers higher accuracy because it scans and analyzes the actual content displayed to users [82].

Hybrid approaches combine URL-based and content-based features to improve the accuracy of phishing detection systems [83]. By leveraging the strengths of both approaches, hybrid models can analyze a wider range of indicators and improve detection rates. These methods typically integrate various ML and DL models to process and classify data from URLs and web page content [84].

## 3.3 | Literature Review and Related Works

Detecting phishing websites is an important step toward combating online fraud. Therefore, it is an attractive research issue that has received extensive research in various studies. In the following, we show you how to extract discriminating features using the most recent classification techniques. A word cloud was created from the abstracts of the selected papers to provide a basic thematic overview of the studies. The visualization



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identifies the most common terms and can help recognize the main research themes in the literature that has been examined, as shown in Figure 13.

### 3.3.1 | Overview of the Selected Studies

The studies included in the literature review were categorized based on the datasets and data sources they have used, the phishing detection methods they have employed, and the metrics they have reported to provide a general overview of the selected literature. The resulting distributions are shown in Figure 14.

The analysis revealed that many public and self-collected datasets are used for phishing detection research. Some of the most popular datasets are used for traditional ML, CNNs, recurrent architectures, and hybrid models. The most frequently reported measures of accuracy are accuracy and F1 score, followed by recall and precision.

### 3.3.2 | ML Techniques

Throughout the last decade, experts in the cybersecurity sector have paid considerable attention and focus to phishing detection based on ML. Substantial research has been undertaken to examine the use of ML in several types of systems to recognize developing phishing attacks. A comparative summary of these studies is presented in Table 5.

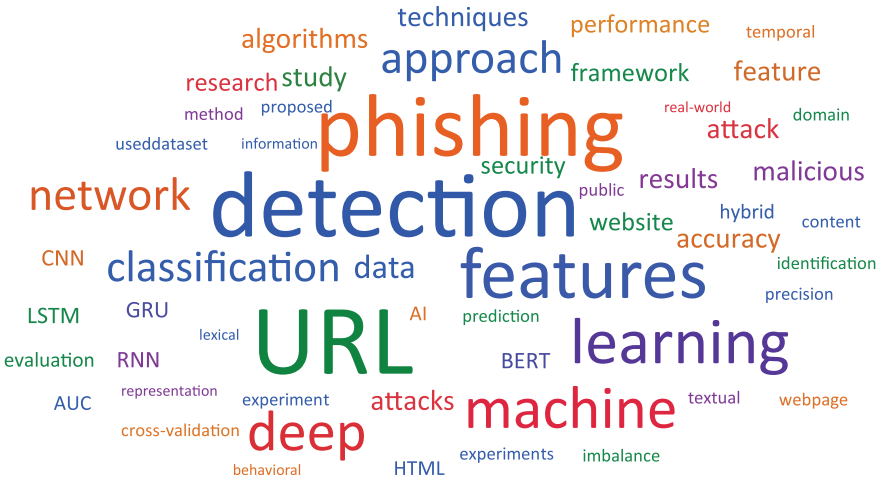



**FIGURE 13** | Word cloud of abstract.

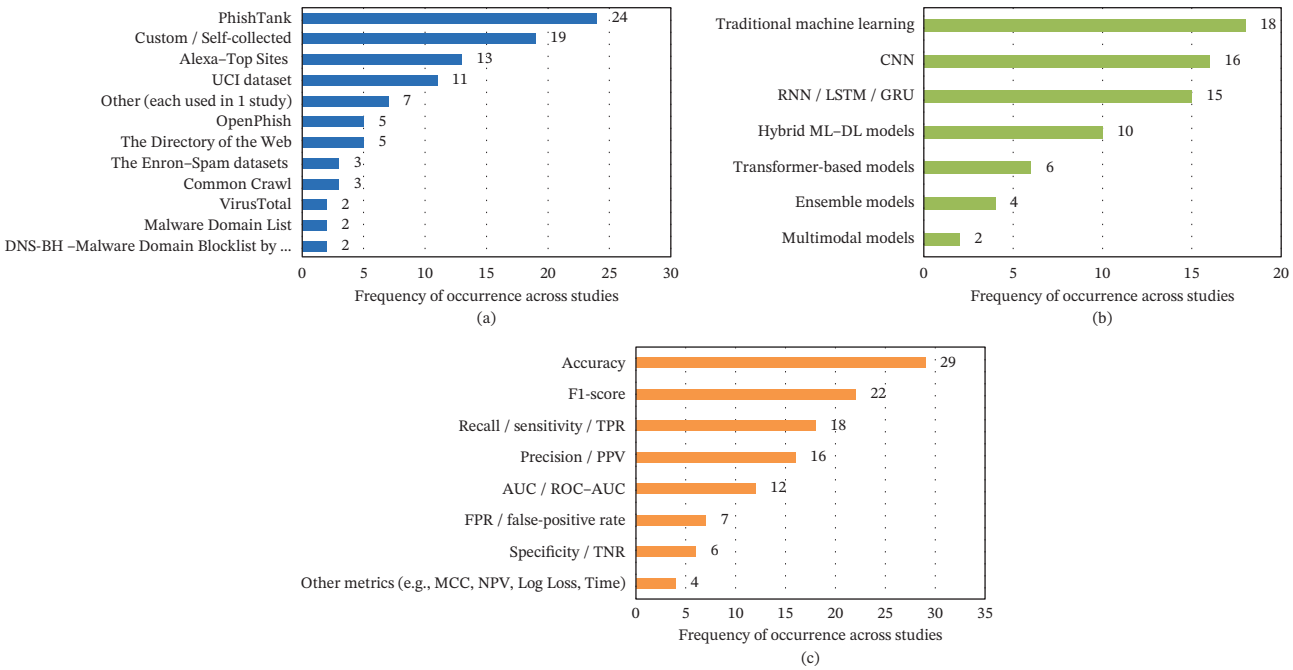



**FIGURE 14** | Descriptive distributions of the datasets, detection approaches, and evaluation metrics reported in the selected studies. (a) Distribution of datasets and data sources used in the selected studies. (b) Distribution of phishing detection approaches used in the selected studies. (c) Distribution of evaluation metrics reported in the selected studies.



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**TABLE 5** | Comparative analysis of machine learning–based phishing detection studies.

| Study | Year | Dataset | Dataset size | Best classifier | Feature selection/representation | Validation protocol | Accuracy/F1 | Limitation and recommended improvement |
|---|---|---|---|---|---|---|---|---|
| Chiew et al. [65] | 2019 | Self-collected phishing dataset; UCI dataset for additional evaluation | 10,000; 11,055 websites | Random forest | HEFS using the CDF-g algorithm and ensemble filter measures | Ten stratified partitions with a 70/30 train–test split in each partition; results were averaged | Acc: 94.60% using 10 features; full-feature RF: 96.17% | The classifiers were evaluated using default WEKA parameters, while the main data were collected between 2015 and 2017. Model tuning and temporally separated validation using recent phishing samples are recommended. |
| Sánchez-Paniagua et al. [69] | 2022 | PILU-90K; PLU-60K login URL subset | 90,000 URLs; 60,000 in PLU-60K | Logistic regression | Character-level TF-IDF n-grams; no explicit feature selection method | 10-fold cross-validation | Acc: 96.50% on PLU-60K; F1: NR | The approach relied only on URL information and showed performance degradation when evaluated on newer URLs. Web page content and visual features should be incorporated, while the training data should be updated periodically. |
| Mughaid et al. [84] | 2022 | Three phishing-email datasets | 16,751; 10,000; 3000 emails | Boosted decision tree; averaged perceptron and neural network for the text dataset | Different predefined feature sets; no explicit feature selection method | 70/30 holdout split | Acc: 88.82%–100%; F1: 88.74%–100% | Different datasets and feature spaces were used across the experiments, reducing direct comparability. A unified experimental protocol, stratified cross-validation, and external testing are recommended. |
| Das Guptta et al. [83] | 2022 | PhishTank phishing URLs and UNB legitimate URLs | 6000 websites | XGBoost | Hybrid URL-based and hyperlink-based features | 80/20 holdout split | Acc: 99.17%; F1: 99.15% | The relatively small and balanced dataset may not represent real-world class distributions. Moreover, DOM-based features may be affected by embedded or obfuscated web page content. Larger temporal datasets and robust content or visual representations are recommended. |

(Continues)



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**TABLE 5** | (Continued)

| Study | Year | Dataset | Dataset size | Best classifier | Feature selection/representation | Validation protocol | Accuracy/F1 | Limitation and recommended improvement |
|---|---|---|---|---|---|---|---|---|
| Adane et al. [85] | 2024 | DS-1 and DS-2 phishing website datasets | 11,430; 11,054 websites | CatBoost | UFS, RFE, MI, and PCC; the best combination was CatBoost-UFS | 80/20 on DS-1 and 70/30 on DS-2, with additional cross-validation comparisons | DS-1: Acc 97.64%, F1 97.62%; DS-2: Acc 97.20%, F1 97.48% | Although testing on a second dataset strengthened the evaluation, the approach remained dependent on predefined handcrafted attributes. Deep or hybrid feature representations and temporal external validation are recommended. |
| Alazaidah [86] | 2024 | Two UCI phishing website datasets | 11,055; 1353 instances | FilteredClassifier and J48 on Dataset 1; random forest on Dataset 2 | Four feature selection methods; InfoGainAttributeEval achieved the best results | Validation protocol was not clearly reported | DS1: Acc 90.76%, F1 90.8%; DS2: Acc 97.26%, F1 97.3% | All classifiers were evaluated using default WEKA settings, the validation protocol was unclear, and feature selection was evaluated only on Dataset 1. Nested cross-validation, model tuning, and external validation are recommended. |
| Sawe et al. [87] | 2024 | Kaggle spam mails dataset | 5171 emails | K-nearest neighbors | Sentence-level Word2Vec representations extracted from email subjects and bodies | 80/20 holdout split | Acc: 97%; F1: NR | The dataset contained spam and ham labels rather than verified phishing and legitimate labels. URLs and email addresses were also excluded. Phishing-specific datasets, cross-validation, URL and header features, and contextual language representations are recommended. |

(Continues)



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**TABLE 5** | (Continued)

| Study | Year | Dataset | Dataset size | Best classifier | Feature selection/representation | Validation protocol | Accuracy/F1 | Limitation and recommended improvement |
|---|---|---|---|---|---|---|---|---|
| Javeed et al. [88] | 2025 | UCI Phishing Websites dataset | 11,055 instances; 30 features | Random forest | Feature-importance and correlation analysis; the final selected subset was not clearly specified | Stratified holdout with 5-fold cross-validation for hyperparameter tuning | Acc: 97.7%; phishing F1: 98% | The evaluation was restricted to one static benchmark dataset, while the proposed hybridization was not sufficiently defined. Evaluation on recent external datasets and a reproducible hybrid and feature selection procedure are recommended. |

Sánchez et al. [69] studied phishing URL detection in a more realistic scenario by considering legitimate login URLs instead of legitimate classes only through home pages. The authors used the PILU-90K dataset to evaluate three different approaches: handcrafted ML techniques, character-level CNN models, and character n-grams (TF-IDF) with LR, finding that the latter outperformed the other two with an accuracy of 96.50% for the login URL set. However, the results revealed that models trained with legitimate homepage URLs incorrectly classified over 30% of the legitimate login URLs, which means that good accuracy on a conventional dataset does not necessarily guarantee reliability on a real one. The study also showed that the performance drops when the models are trained with older datasets and tested with more recent URLs, which indicates the significance of freshness and class representation of datasets. However, the analysis focused only on information from URLs and not on content or visual information in web pages, which were reserved for future research.

Mughaid et al. [84] compared seven ML phishing-email detection classifiers with three different sets of phishing-email feature representations. The training and testing of the datasets were performed as 70% training and 30% testing, with the boosted DT having an accuracy of 88.82% and 100% on the first two datasets, respectively; the neural network having an accuracy of 97.7% on the text-based dataset, and the averaged perceptron an accuracy of 88.82% on the third dataset. The comparison showed that the number of features affects the accuracy of the classification, but did not show any direct comparison of the models, and it was only tested on one train–test split and on different datasets for the experiments, so generalization to a never-before-seen phishing source was not proven.

Das Guptta et al. [83] suggested a phishing website detection solution based on both URL-based and hyperlink-based features, which does not require search engines or third-party websites. The hybrid representation was initially designed with 25 features created by hand and tested with five ML classifiers, with XGBoost showing the highest accuracy of 99.17%. A balanced sample set of 6000 websites and a single (80/20) train–test split was used; however, this may limit the generalizability of the reported performance to larger and growing real-world datasets.

Adane et al.'s [85] four feature selection techniques, CatBoost, Gradient Boosting, RF, and LR, were compared with each other. To test the consistency of the best model on different sets of features, two datasets of phishing websites were used, with 11,430 and 11,054 samples, respectively. The best performance was demonstrated by CatBoost and Univariate Feature Selection, which achieved an accuracy of 97.64% and an F1 score of 97.62% on DS-1 and 97.20% accuracy and an F1 score of 97.48% on DS-2. While validation in a second dataset is valuable, the study only considers predefined features and traditional ML models and does not evaluate the performance of DL or temporal robustness to changing phishing attacks.

Chiew et al. [65] proposed an automatic phishing website detection feature selection framework called hybrid ensemble feature selection (HEFS). The framework was evaluated using six ML classifiers and the CDF-g algorithm with data-perturbation ensembles and function-perturbation ensembles. By using only 10 of the original 48 features, the RF model missed just 0.60% of the accuracy of the model using all features and reduced the



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feature dimensionality by 79.2%. The framework was also tested on a UCI dataset; however, the classifiers used were tested with default parameters, and the main dataset was gathered from 2015 to 2017, which could restrict the relevance of the selected baseline features to current phishing patterns.

Javeed et al. [88] used the UCI phishing website dataset (11,055 instances, 30 predefined features) for phishing URL detection using DTs, RF, C-Support Vector Classification, and AdaBoost. Fivefold cross-validation was used for hyperparameter tuning, with the remaining samples forming the test set used for the final evaluation. RF was the most successful, with 97.7% accuracy, an AUC of 99.7%, and a phishing-class F1 score of 98%. However, the evaluation was based on a single benchmark set of phishing sites using a set of handcrafted features and was not validated against external or temporal phishing sets, thus restricting the evidence of generalization to the appearance of new phishing sets in the benchmark set.

Alazaidah et al. [86] evaluated 24 different ML classifiers based on six learning strategies with two phishing datasets in the UCI repository. FilteredClassifier and J48 showed the best performance on the binary dataset with 90.76% accuracy and 90.8% F1 score, respectively, while RF performed the best on the multiclass dataset with 97.26% accuracy and 97.3% F1 score. In addition, four feature selection methods were compared, and InfoGainAttributeEval gave the best overall performance. The validation method was not clearly described; all classifiers were applied with their default settings from WEKA, and only one of the datasets was used for the feature selection analysis, which restricted the validity of the general statements.

Sawe et al. [87] proposed a model for detecting phishing emails at the sentence level based on the Word2Vec word representation and the KNN classifier approach. The model was tested against the 80/20 split of 5171 emails of spam and ham from a Kaggle dataset and evaluated on the text content of the email subject and body. The proposed model obtained a testing accuracy of 97% and a training accuracy of 99%. However, the dataset was not labeled as phishing or trusted email but as spam and ham, thus lowering the direct validity of the results for phishing detection. Furthermore, there was no external validation, and only one holdout split was used for the evaluation.

### 3.3.3 | DL Techniques

In particular, phishing detection has been gaining momentum in DL for its ability to learn complex patterns from data. Several approaches have been proposed that use DL to improve the accuracy of phishing detection against evolving phishing attacks. A comparative summary of these studies is presented in Table 6.

Aljofey et al. [89] designed a CNN to automatically learn sequential patterns directly from the URL strings without downloading a website or relying on any third-party services at the character level. The four feature representations were character embeddings, character-level TF-IDF, count vectors, and handcrafted features. The model was tested with four sets of URLs with 318,642, 73,575, 83,857, and 82,888 URLs, respectively, without giving a ratio for the train–test split. The character-level CNN model showed the best results on D2, with an accuracy of 98.58% and an F1 score of 98.56%. Although tested with several datasets, it was found that only URL strings were used, and that the method was relatively time-consuming to train; short URLs or those with sensitive terms were misclassified. It was recommended to incorporate web page code and text in future work.

Aldakheel et al. [90] proposed a CNN with a single layer, which can classify a URL as phishing or legal. The study employed a balanced sample of 20,000 URLs (10,000 phishing URLs taken from PhishTank and 10,000 legitimate URLs). Seven URL-based features were extracted, and the data were split into 70:30 training testing sets. The proposed model achieved an accuracy of 98.1%, a macro-F1 score of 97.9%, and a micro-F1 score of 98.1%. However, this evaluation was only performed on a single balanced dataset, and external and temporal validation was not performed. Further research is needed to test the model with larger, more varied, and temporally more recent data, to test the model against datasets from external sources, and to explore other features based on URL, content, and domain to enhance generalization and robustness to changing phishing attacks.

Alshingiti et al. [91] compared CNN, LSTM, and LSTM–CNN models for classifying phishing and legitimate URLs. The experiments used 20,000 URL records with 80 initial features, of which the 30 highest-ranked features were selected using SelectKBest and normalized using MinMax scaling. The dataset was divided into 80% for training and 20% for testing. CNN achieved the best performance, with 99.2% accuracy, 99% precision, 99.2% recall, and a 99.2% F1 score. However, the evaluation relied on a single static dataset and did not verify whether the assessed websites were still active. Recommendations for reducing training time, improving feature engineering, and incorporating web page context alongside URL-based features.

Adebowale et al. [51] developed an intelligent phishing detection system (IPDS), a hybrid DL system that combines LSTM to process URL, textual, and frame information with a CNN to process web page images. The framework was trained by combining over 10,000 images from legitimate sites and over 10,000 images from phishing sites, provided by Common Crawl and PhishTank, respectively, with one million balanced URLs. The authors tested holdout validation and 2-, 5-, and 10-fold validation, with 70/30 holdout validation giving the best performance. The IPDS achieved 93.28% accuracy, 93.30% precision, 93.27% recall, and a 93.29% F-measure. The multimodal design may be more useful in detecting phishing activity than URL-only designs, but its average detection time is approximately 25 s, which might constrain its use in real-time browser use. Furthermore, an external dataset separate from the evaluation was not used. It is therefore recommended that faster inference and external validation be used.

Alsubaei et al. [55] proposed RNT-J, a hybrid phishing detection framework based on ResNeXt, GRU, and an attention mechanism, which was optimized using the Jaya algorithm. The framework is an ensemble of autoencoder and ResNet representations (EARN) to extract features, and it uses SMOTE to resolve the class imbalance problem. The experiments were conducted using a Kaggle phishing website dataset, which has 10,000 URL and web page instances, out of which 3.27% of the instances are phishing samples. When trained with 75% of the data and tested with 25% of the data, RNT-J yielded 98% accuracy, 97.8% precision, 99% recall, 98.8% F1 score, and 99.8% AUC, which outperformed the compared SVM, CNN, DenseNet, ResNet, BERT, LSTM, and RNN models. However, the validation



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**TABLE 6** | Comparative analysis of deep learning–based phishing detection studies.

| Study | Year | Dataset | Dataset size | Best model | Feature selection/ representation | Validation protocol | Accuracy/F1 | Limitation and recommended improvement |
|---|---|---|---|---|---|---|---|---|
| Aljofey et al. [89] | 2020 | Four URL datasets (D1–D4) collected from PhishTank, Alexa, Yandex, Common Crawl, and other sources | 318,642; 73,575; 83,857; and 82,888 URLs | Character-level CNN | Character embeddings; compared with character-level TF-IDF, count vectors, and handcrafted features | Random train–test split; the splitting ratio was not clearly reported | Best on D2: Accuracy 98.58%; F1 98.56% | The method relied only on URL strings, required relatively long training time, and could misclassify short URLs or URLs containing sensitive terms. Web page code and textual features should be incorporated. |
| Adebowale et al. [51] | 2023 | PhishTank, Common Crawl, and collected web page images | 1 million URLs, equally divided between phishing and legitimate samples; more than 10,000 images | IPDS: LSTM + CNN | Hybrid URL, text, frame, and image representations; AlexNet-based image analysis | 70/30 holdout; additionally compared with 2-, 5-, and 10-fold cross-validation | Accuracy 93.28%; precision 93.30%; recall 93.27%; F1 93.29% | The average detection time was approximately 25 s, which may restrict practical browser deployment. No independent temporal dataset was used; faster inference and external validation are recommended. |
| Aldakheel et al. [90] | 2023 | PhishTank phishing URLs and collected legitimate URLs | 20,000 URLs: 10,000 phishing and 10,000 legitimate | 1D-CNN | Character-level URL embeddings | 70/30 holdout split | Accuracy 98.10%; macro-F1 97.90%; micro-F1 98.10% | The evaluation was restricted to URL-based samples from one dataset. The reported 98.77% value represented training accuracy rather than test accuracy. External, temporal, and multimodal evaluations are recommended. |

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**TABLE 6** | (Continued)

| Study | Year | Dataset | Dataset size | Best model | Feature selection/ representation | Validation protocol | Accuracy/F1 | Limitation and recommended improvement |
|---|---|---|---|---|---|---|---|---|
| Alshingiti et al. [91] | 2023 | ISCX-URL2016 | 20,000 URLs; 80 initial features | CNN | SelectKBest reduced the original 80 features to 30; MinMax scaling was applied | 80/20 holdout split | Accuracy 99.20%; precision 99.00%; recall 99.20%; F1 99.20% | The study relied on one static dataset and did not verify whether the evaluated URLs were still active. Recent external datasets and web page-context features should be included. |
| Alsubaei et al. [55] | 2024 | Kaggle phishing website dataset | 10,000 URL and web page instances; phishing represented 3.27% before balancing | RNT-J: ResNeXt-GRU with attention and Jaya optimization | SMOTE; EARN feature ensemble using Autoencoder and ResNet representations; PCA | 75/25 holdout split | Accuracy 98.00%; precision 97.80%; recall 99.00%; F1 98.80%; AUC 99.80% | The evaluation relied on one dataset without external or temporal testing. The workflow appears to apply SMOTE before data splitting, which may cause leakage. Oversampling should be applied only to the training subset. |
| Zhu et al. [54] | 2024 | DATA1 and DATA2 | DATA1: 10,000 samples (5000 phishing and 5000 legitimate); DATA2: 1,172,577 URLs (587,668 phishing and 584,909 legitimate) | PDHF: CNN–DQA deep features combined with random forest | InVa feature ranking; simulated-annealing-based bidirectional search; one-dimensional character CNN with disorderly quantized attention | DATA1: 10-fold cross-validation; DATA2: 80% training, 10% validation, and 10% testing | Accuracy 99.65%; precision 99.42%; recall 99.40%; F1 99.41% | The model performed well on two large datasets using handcrafted and deep URL features. However, it was not independently validated across time and datasets, and its deep features covered only URL strings; therefore, external, temporal, and multimodal testing is recommended. |

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**TABLE 6** | (Continued)

| Study | Year | Dataset | Dataset size | Best model | Feature selection/ representation | Validation protocol | Accuracy/F1 | Limitation and recommended improvement |
|---|---|---|---|---|---|---|---|---|
| Sahingoz et al. [92] | 2024 | Self-collected multi-source URL dataset released with DEPHIDES | 5.1M URLs: 2.3M phishing and 2.8M legitimate | CNN | Character-level embeddings with a maximum URL length of 200 characters | Train, validation, and test sets; architectural configurations were initially evaluated using a 10% subset | Accuracy 98.74%; aggregate F1 was not clearly summarized | Most architectural comparisons were performed using the 10% subset because of hardware constraints. All architectures should be evaluated on the complete dataset and tested using independently collected and temporally separated URLs. |
| Zhou et al. [93] | 2024 | Kaggle phishing-email dataset | Total size: NR; test set: 3508 emails | MobileBERT with CMA-ES optimization | Email cleaning and tokenization; concatenation and averaging of the final four MobileBERT hidden layers; CMA-ES hyperparameter optimization | Train–test split and validation protocol were not clearly reported | Accuracy 95%; weighted F1 95%; phishing precision 97%; recall 95%; F1 96% | The evaluation relied on one dataset, while the total dataset size, data split, final classification architecture, and ablation protocol were insufficiently reported. Clearer methodological reporting and validation on larger, external, and temporally diverse datasets are recommended. |
| Barik et al. [94] | 2025 | Primary dataset from Kaggle, Alexa, and PhishStorm; secondary PhishTank dataset | Primary: 27,534 balanced URLs; secondary: 20,004 balanced URLs | EGSO-CNN | StandardScaler, TF-IDF, VAE feature extraction, and Enhanced Grid Search Optimization | 80/20 holdout; additional validation using the secondary dataset | Primary: Accuracy 99.44%, F1 99.32%; secondary: Accuracy 99.40%, F1 99.25% | The model requires evaluation on more datasets and cannot detect web pages that replace textual content with embedded Flash or JavaScript objects. Temporal testing, embedded-object analysis, and transfer learning are recommended. |



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was external, and temporally independent data were not used in the evaluation. Furthermore, the illustrated pipeline uses SMOTE before splitting the data, which may lead to data leakage because synthetic samples could be created before the splitting of the test set. External validation and only training oversampling are thus necessary to confirm the claimed generalization.

Zhu et al. [54] introduced a hybrid phishing detection model (Pdhf) by combining some handcrafted features with automatically learned deep URL representations. Using the Information Value (InVa) index and simulated-annealing-based bidirectional search, the initial 48 URL- and HTML-based features were pruned to 18 features, and CNN–DQA was used to learn complementary URL features. Tenfold cross-validation was used on DATA1, which consisted of 10,000 balanced samples, and a train–validation–test split with 80/10/10 on DATA2, which consisted of 1,172,577 URLs. PDHF achieved 99.65% accuracy and a 99.41% F1 score. No independent cross-dataset and temporal validation was performed; however, the deep features were extracted from URL strings instead of web pages or HTML.

Sahingoz et al. [92] introduced DEPHIDES, a URL-based phishing detection system that evaluates ANN, CNN, RNN, bidirectional RNN, and attention-based architectures with character-level representations. The authors created a massive dataset of approximately 5.1 million URLs, of which 2.3 million were phishing URLs and 2.8 million were legitimate URLs, and additionally a 10% subset that was more suitable for experiments with less computational power. CNN had the highest test accuracy of 98.74% and could classify 130,070 URLs per second. Most architectural comparisons were performed on the smaller subset owing to hardware restrictions, and there was no external dataset collected or temporally separated to report evaluation on. An independent external test of both architectures with the entire dataset would further support the results.

Zhou et al. [93] proposed an optimized phishing-email detection framework for Semantic Web systems based on a MobileBERT-based contextual representation and Covariance Matrix Adaptation Evolution Strategy (CMA-ES). Stop words and unwanted characters were removed from the text in the email, and the email text was subsequently tokenized; the last four hidden layers in MobileBERT were concatenated and averaged to obtain semantic feature vectors. The parameters, including learning rate, number of layers, batch size, and activation functions, were then optimized using CMA-ES. An overall accuracy and weighted F1 score of 95% were obtained with the model. In the phishing class, it achieved the highest accuracy (97%), recall (95%), and F1 score (96%). While the results show that optimized transformer representations work well, the study was tested with only one Kaggle email dataset, and the size of the dataset, the division between the training and test sets, and the validation procedure were not well documented. Moreover, the final classification architecture was not well described, which also hampered the possibility of reproducibility. Therefore, there is a need to evaluate larger and diverse external datasets with clear ablation and validation protocols.

Barik et al. [94] evaluated optCnnPhishingApe: An optimized DL model for phishing URL detection using the component of a standard scaler in preprocessing, variational autoencoders to extract features, and enhanced grid search optimization to tune the hyperparameters of the CNN model. The final primary dataset consisted of 27,534 balanced samples, and a secondary PhishTank dataset with 20,004 balanced URLs was employed for further validation. The proposed model yielded 99.44% accuracy, 99.98% precision, 99.21% recall, and a 99.32% F1 score on an 80/20 train–test split for the first dataset and 99.40% accuracy and 99.25% F1 score for the second dataset. The authors did note, however, that the model needs to be tested with other sets of pages and cannot recognize phishing pages that change the text with inline Flash or JavaScript objects. To evaluate the aforementioned issues, temporally collected datasets, embedded content analysis, transfer learning, and user-feedback assessments should be included in future evaluations.

The reviewed studies are summarized to emphasize the key performance characteristics, variations in methodology, and issues with the strength and broader applicability of phishing detection models. The features of the datasets and the evaluation protocol used in the reviewed studies were found to be important factors influencing reported performance; therefore, high accuracy should not be taken as evidence of model robustness. Small, balanced, and/or randomly split datasets can produce very high accuracy; however, larger and more heterogeneous datasets, external datasets, and datasets that are temporally separated can yield better evidence of generalizability. The incorporation of informative handcrafted URL and web page features continues to be useful with ML approaches, while DL approaches can learn more complex representations with fewer manual features and often have better performance on a smaller number of examples and in lower dimensions; however, they generally require more resources, such as more data and higher computing power. The comparison of different studies is also limited by differences in the preprocessing, class distribution, representation of the features, type of validation, and type of evaluation metrics used, as well as the presence of duplicated or overlapping samples. The results indicate that future phishing detection research should focus on dataset quality, leakage-aware evaluation, temporal robustness, external validation, and computational feasibility, in addition to the reported accuracy, as these aspects are crucial to improving detection.

### 3.4 | Key Challenges in Phishing Detection

The analyzed research showed that there are recurring challenges in existing phishing detection techniques. These problems are driven by the ever-changing nature of phishing attacks, the availability of limited data and phishing evaluation methods, and the challenge of providing stable performance in operational settings. The main challenges are summarized as follows:

1. Evolving and Sophisticated Phishing Techniques: Phishing methods are continually improved and evolve to become more convincing and more evasive, such as domain impersonation, URL obfuscation, visually deceptive web pages, dynamically generated content, and messages created using AI. These new techniques make static rules and models built from previous attack patterns less effective [30, 94].
2. Feature Extraction and Representation: It is difficult to extract reliable features from URLs, the text of emails, HTML code, web page information, screenshots, and domain information. Phishing sites can include foiled elements in images, scripts, shortened URLs, redirections,



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or dynamically generated elements on any web page. As a result, techniques that rely on one type of feature may not find all the evidence of phishing [51, 52, 89].

3. Dataset Limitations: Many studies utilize small, static, balanced, and/or outdated datasets that may not be representative of current phishing campaigns in the wild. Direct comparisons between studies are limited by variations in the sources of data, period of study, definitions of classes, and features included. Old datasets might not include phishing samples of the same type that are found in newer samples. Phishing samples could be different types in old datasets than in newer samples [53, 65, 68].
4. Class Imbalance: Phishing real-world datasets are often very imbalanced and have a significant number of legitimate samples compared to phishing samples. The models trained on the above-mentioned datasets can have a high level of accuracy and low recall for the phishing class or a low F1 score for the phishing class. Thus, it is important to provide class-specific metrics, as well as the overall accuracy [30, 55].
5. Duplicate Samples and Data Leakage: The same or similar URLs can be present in both the training and testing sets, especially if data are gathered from the same or similar sources. This can result in some data leakage and artificially high-performance estimates. Therefore, before the process of model training and evaluation, it is essential to conduct deduplication, source-aware partitioning, and explicit leakage checking [95].
6. Temporal Drift: The targeted URLs, page layout, and attack approaches evolve. Randomly partitioning samples collected at the same time may not fairly evaluate a model's predictive capability. Temporally separated training and testing sets are required to assess long-term robustness [72].
7. Limited Generalization and External Validation: Several proposed models have been tested using one dataset or one random train–test split. Thus, their generalization capabilities to other phishing sources, new campaigns, and real-life scenarios are still uncertain. To support greater evidence of generalizability, independent cross-dataset evaluation is needed [89, 91, 93, 94].
8. Real-Time Detection: A phishing detection system should be accurate, fast, and cheap to compute. Models that need to download to the web page, process images, make a WHOIS query, or use external services or complex DL models can cause delays that make them less suitable for deployment in a browser or in an email security device [51, 92, 96].
9. Evasion and Bypass Attempts: Attackers can alter the URL, page design, text, or scripts, or any visual features of the page to avoid detection. These types of systems that depend on one modality are especially prone to such maneuvers. More adaptive and multimodal strategies are needed to decrease the impact of evasion strategies [52, 89, 94].
10. False Positives and False Negatives: True positives can be misclassified as phishing due to the similarities that exist among legitimate login pages, newly registered domains, and shortened URLs. However, a more advanced phishing page that closely resembles a legitimate site could be misidentified as a legitimate site. Minimizing both errors is a significant challenge for implementation [90, 96, 97].
11. Influence Techniques: As a result of current developments, phishing emails use a variety of persuasive techniques to convince victims to respond, making them difficult to detect using static or dynamic analysis alone [30].
12. Diverse Attack Vectors: Phishing attacks can be sent via email, websites, social media, SMS text messages, QR codes, cloud platforms, collaboration applications, and more. Each channel has its own set of indicators; therefore, one detection method is unlikely to provide a complete solution for phishing detection [14, 30].

In this context, the need for phishing detection systems using recent and diversified datasets, leakage-aware and temporally separated evaluation protocols, multimodal feature representations, external validation, and efficient models that can maintain low false-positive rates and respond to "real-time" and continuously evolving phishing attacks is raised.

### 3.5 | Future Works

Considering the limitations and methodological gaps of the reviewed studies, future studies should focus on the following directions:

1. Developing Diverse and Temporally Updated Datasets: Much research focuses on the continuous improvement and development of ML and DL models to enhance their accuracy and efficiency. For example, some studies suggest that future research could explore incorporating additional features and using more sophisticated algorithms to improve recognition rates, as well as developing hybrid frameworks to enhance their applicability and robustness in real-world situations [38, 45, 77, 84].
2. Improving Dataset Quality and Evaluation Reliability: Duplication removal, conflicting labels, class imbalance, and data leakage should be taken into account more. It is recommended that future studies use leakage-aware data splitting [54], temporal evaluation, cross-dataset testing, and external validation to provide more reliable evidence for the performance of the models on unseen phishing attacks [53, 96].
3. Enhancing Real-Time Detection Capabilities: Future research should focus on creating lightweight, scalable, and efficient models that are feasible to use in the browser, email, mobile, and enterprise security environments. In addition to accuracy, detection speed needs to be considered, along with robustness and resources [44, 51].
4. Reducing False Positives and Improving Robustness: Further research is needed to reduce false-positive rates, especially for brand-identity and web page-similarity detection techniques. Additionally, it is important to test the models with adversarial manipulation, obfuscated URLs, zero-day attacks, and concept drift so that the model is useful with the evolution of phishing methods [52].



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5. Advancing Feature Representation and Hybrid Detection: Future work requires further investigation of enhanced feature representation and extraction techniques that integrate URL, HTML, textual, visual, domain, and behavioral features [42]. Hybrid approaches could yield more comprehensive phishing detection, as could models based on character-level representations, NLP, and transformer-based models with careful control of model complexity [74].

6. Integrating User Behavior and Awareness: User behavior analysis is an additional technical detection system that can be integrated to enhance it, including user interactions with emails, links, and web pages. Concurrently, awareness and education initiatives must be created to educate users about the new phishing techniques and how to deal with suspicious messages [75, 84].

7. Exploring Emerging AI Paradigms: Future work should consider the usage of LLMs and VLMs to analyze phishing in context and the usage of GNNs to model the relationship between URLs and domains, and federated and continual learning to ensure privacy-preserving and adaptive detection. To enhance interpretability and robustness against phishing attacks from AI-generated and evasive emails, XAI and adversarial evaluation are required.

## 4 | Conclusions

This systematic literature review examined phishing detection studies published between 2019 and 2025, focusing on ML, DL, transformer-based, and hybrid approaches. The reviewed studies show clear progress in terms of classification performance, automated feature learning, and real-time detection. However, the reliability and practical applicability of many proposed systems remain limited by outdated or insufficiently diverse datasets, class imbalance, duplicate samples, inconsistent data splitting, potential data leakage, and limited external or temporal validation datasets. Differences in datasets, class distributions, evaluation protocols, and reported metrics also make direct comparisons between studies difficult. Overall, high accuracy alone is insufficient to demonstrate real-world effectiveness; evaluations should also consider generalizability, robustness, detection speed, computational cost, and resistance to evolving attacks. Therefore, future studies should prioritize updated datasets, leakage-aware evaluations, cross-dataset testing, lightweight real-time models, and robust hybrid representations to support more reliable and scalable phishing detection systems.


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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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